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# Course Orientation & Data Science Process

What data science is and how a project runs

**Prof. Gabriel Spadon, PhD**
Faculty of Computer Science
spadon@dal.ca

# How and Where We Meet

- Tue and Thu, **16:05–17:25**.
- Carleton Tupper Building, #**3H01**.
- Term runs for **twelve teaching weeks**; see the official calendar.
- Integrated labs begin in Thursday classes; there is no separate lab section.
- Classes mix ungraded questions, worked examples, and pair work.

## Shared Expectations

- Complete your work on time.
- Respect other people's space.
- Bring your questions to class.
- Communicate respectfully.

# Today's Data Science (DS) Questions

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### Why Data Science? 1

Where the demand comes from, and what the work in a data-related job is.

### What Is Data Science? 2

The shapes data arrives in, one clinical table, and a definition to hold them.

### What Is the Process Like? 3

Three kinds of question, and CRISP-DM, the six-stage loop a project runs.

### How Does This Course Work? 4

Labs, assignments, examinations, and where documents are posted.

# Why Learn Data Science?

Demand for data skills, the volume of data, and computing power have grown together.

# A 2022 Job-Market Snapshot

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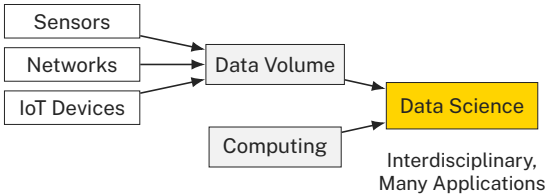
Indeed's 2022 top twenty jobs in Canada cover all fields; two of the twenty are data roles. Growth is in each role's share of job postings. This is not a current salary guide.

| Rank | Job | Average Salary | Share Growth (%), 2019–22 |
|---|---|---|---|
| **1** | **Data engineer** | **$85,706** | **242** |
| 2 | Site reliability engineer | $90,010 | 218 |
| 3 | Engineering manager | $111,515 | 161 |
| … | ranks 4 to 18, fifteen further roles | … | |
| **19** | **Data scientist** | **$76,938** | **60** |

Indeed's twenty best jobs in Canada, 2022, reported by Canadian HR Reporter. hrreporter.com

# What Is Driving the Field

- Increasing volumes of data
  - Variety of cheap sensors
  - Increased networking
  - Internet of Things
- Increasing computing power
- An interdisciplinary field
- A variety of applications



# Application Domains

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### Healthcare

Diagnosis support, risk scoring, and planning.

### Finance

Credit decisions, pricing, and fraud.

### Smart Cities

Mobility, demand, and infrastructure planning.

### E-Commerce

Recommendation and demand forecasting.

### Social Networks

Communities, influence, and information flow.

### Environment

Monitoring, forecasting, and mitigation.

# Data Science by Example

Data of every kind, one table of rows and columns, and a definition of the field.

# Increasing Volumes of Different Types

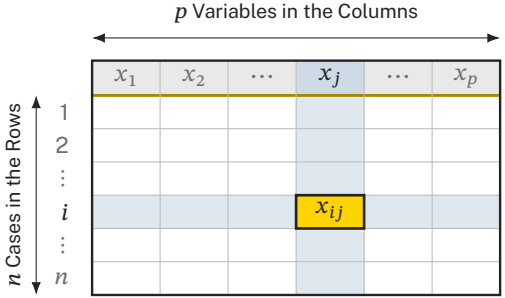
- Tabular Data
- Databases of Related Tables
- Text
- Multimedia: Images, Video, Sound
- Data Streams and Time Series

In This Course:
**Tabular Data**

The other types matter, and the reasoning here transfers to them.

# Tabular Data

- A data set can be represented as an $n \times p$ matrix with $n$ observations or cases, each described by $p$ variables.
- Rows represent the cases, also called objects, examples, instances, or records.
- Columns represent variables, also called features, fields, or attributes.
- The entry $x_{ij}$ is the value of variable $j$ for case $i$. Numeric variables measure quantities; categorical variables record labels.



# The Heart Disease Data Set

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| Row | Age | Sex | ChestPainType | RestingBP | Cholesterol | FastingBS |
|---|---|---|---|---|---|---|
| | numeric | category | category | numeric | numeric | category |
| 1 | 40 | M | ATA | 140 | 289 | 0 |
| 2 | 49 | F | NAP | 160 | 180 | 0 |
| 3 | 37 | M | ATA | 130 | 283 | 0 |
| 4 | 48 | F | ASY | 138 | 214 | 0 |
| 5 | 54 | M | NAP | 150 | 195 | 0 |
| 6 | 39 | M | NAP | 120 | 339 | 0 |

**12 attributes** and **918 cases**. The first six patients, with the first six variables. Types describe their analytical roles. The data set combines five UCI heart-disease databases.

fedesoriano (2021), *Heart Failure Prediction Dataset*, Kaggle. kaggle.com/datasets/fedesoriano

# The Same Patients, More Variables

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| Row | RestingECG | MaxHR | ExerciseAngina | Oldpeak | ST_Slope | HeartDisease |
|---|---|---|---|---|---|---|
| | category | numeric | category | numeric | category | category |
| 1 | Normal | 172 | N | 0.0 | Up | 0 |
| 2 | Normal | 156 | N | 1.0 | Flat | 1 |
| 3 | ST | 98 | N | 0.0 | Up | 0 |
| 4 | Normal | 108 | Y | 1.5 | Flat | 1 |
| 5 | Normal | 122 | N | 0.0 | Up | 0 |
| 6 | Normal | 170 | N | 0.0 | Up | 0 |

Row numbers match the preceding slide. `HeartDisease` is the category to predict, encoded as 0 or 1. Numeric storage does not make a category continuous.

# Reading the Same Table Twice

## R

```r
heart <- read.csv(
  "heart.csv",
  stringsAsFactors = TRUE
)
dim(heart)       # 918  12
head(heart)
str(heart)
table(heart$HeartDisease)
```

## Python

```python
import pandas as pd

heart = pd.read_csv("heart.csv")
print(heart.shape)  # (918, 12)
print(heart.head())
print(heart.dtypes)
heart["HeartDisease"].value_counts()
```

Both calls produce 918 rows and 12 columns. R imports text as factors, its category representation. Python uses pandas, a library for tables; a notebook shows only a cell's last value, so earlier lines use `print`. Storage types and analytical roles can differ.

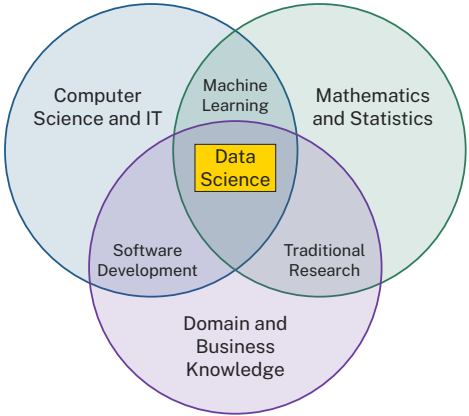
# What Is Data Science?

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"Data mining is the analysis of (often large) observational data sets to find unsuspected relationships and to summarize the data in novel ways that are both understandable and useful to the data owner."

Hand, Mannila, and Smyth, *Principles of Data Mining*, MIT Press, 2001, page 1.

Data science brings computing, statistics, and domain knowledge to this work (right). **Observational** means the researcher does not assign an intervention.

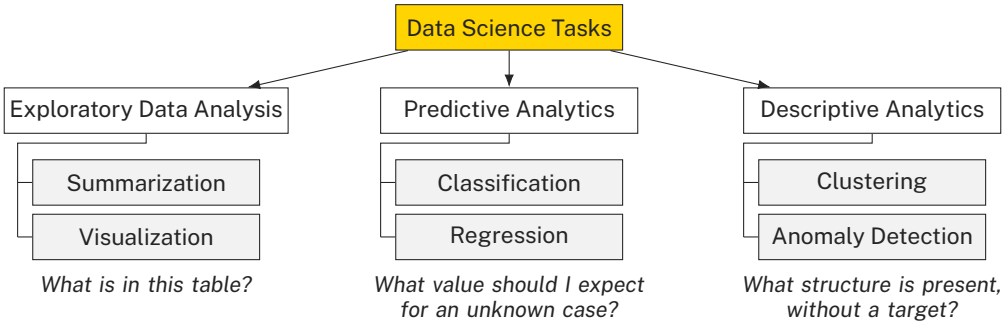



Three-circle framing after Drew Conway (2010); labels are this course's.

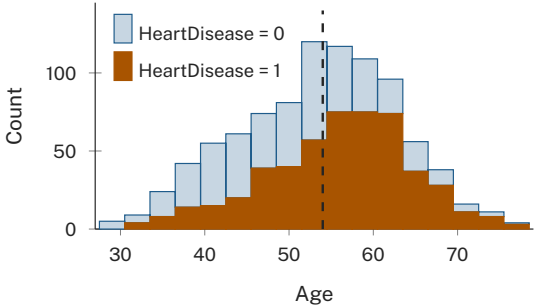
## Data Science Tasks

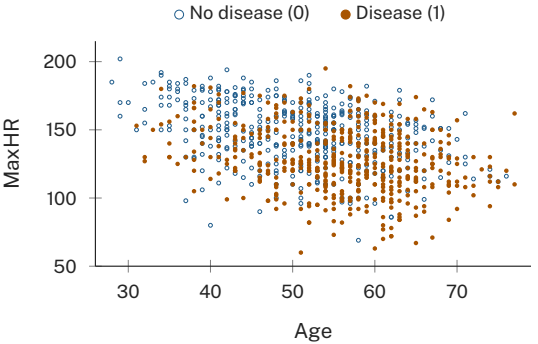
Three common tasks split the work, each one defined by the question it answers.

# Data Science Tasks



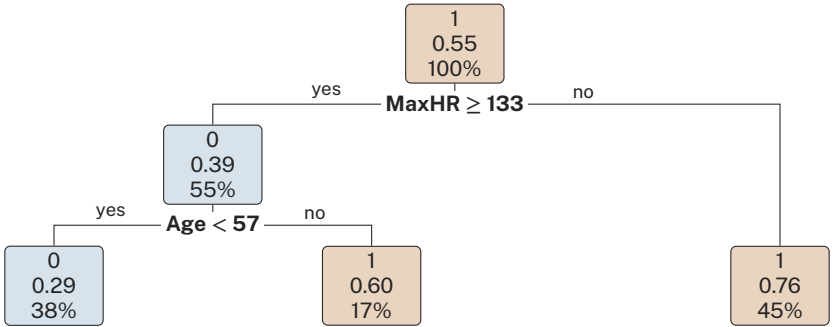
# Exploratory Data Analysis



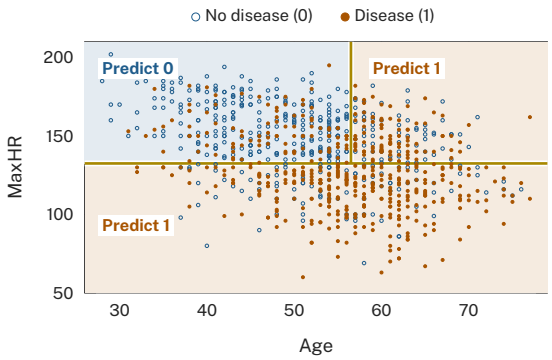

The histogram counts patients in age intervals. The scatter places one patient at each (Age, MaxHR) pair; overlapping points may hide patients. The dashed line marks the median age, 54.

# Predictive Analytics, the Fitted Tree




A classification tree predicts a category by following rules. This one was fitted on Age and MaxHR only, limited to three leaves. Boxes show the prediction, disease proportion, and share of patients used to fit the rules.

# The Partition the Tree Induces




- Two splits cut the plane in three.
- The low-heart-rate region predicts disease here.
- Above the heart-rate threshold, age decides.

**Training accuracy: 71.24%**

654 of 918 training cases. Split boundaries are 132.5 and 56.5, equivalent to the integer rules shown. Future performance is covered in Week 9.

# Descriptive Analytics, an Example




- No target variable is used.
- $k$-means forms three groups from Age and MaxHR, both rescaled to a common scale.
- The five patients farthest from their group's centre are flagged, CSV rows 830, 3, 508, 391, and 569.

Clustering and outlier detection are taught in Weeks 11–12. Today, identify the question and read the highlighted observations.

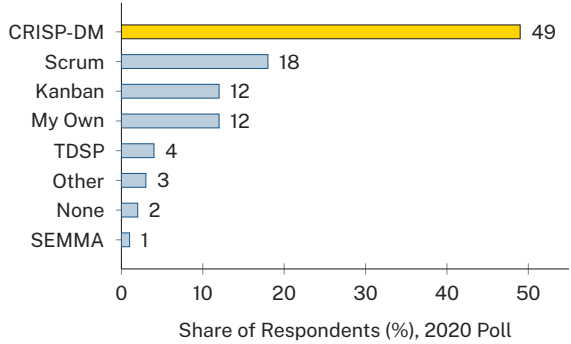
# CRISP-DM, the Typical Data Science Process

Six stages connect a question to a useful result. New evidence can send us back.

# CRISP-DM

- **CR**oss **I**ndustry **S**tandard **P**rocess for **D**ata **M**ining.
- The most selected option in this practitioner poll.

The 2020 convenience poll has 109 respondents. It does not estimate prevalence among all practitioners. Rounded percentages sum to 101%.




Data Science Process Alliance poll, 2020, 109 respondents. datascience-pm.com

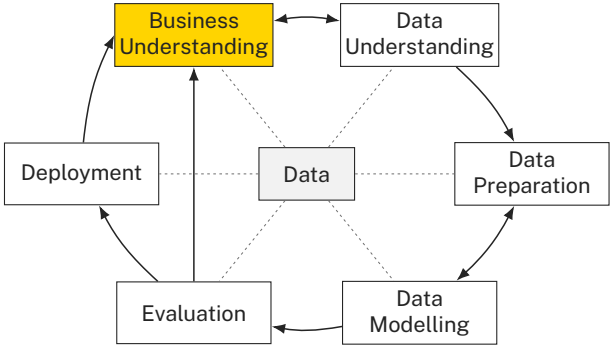
# Business Understanding

## 1. Business Understanding

What are the business goals?

How can data science help achieve them?

Name the decision and who will use the result. Revisit the question if the available data cannot answer it.




Cycle after the CRISP-DM reference model, Chapman et al. (2000).

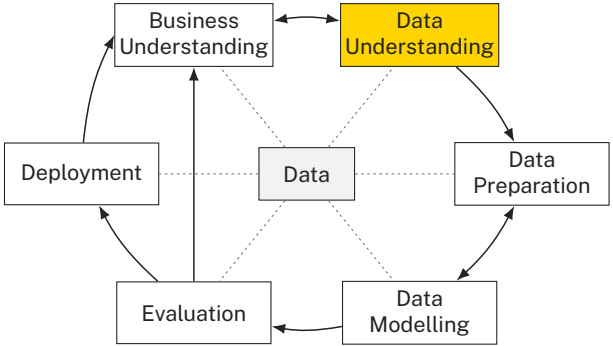
# Data Understanding

## 2. Data Understanding

What data do we have or need?

Are there data quality issues?

*Can you think of any potential data quality issues or limitations?*



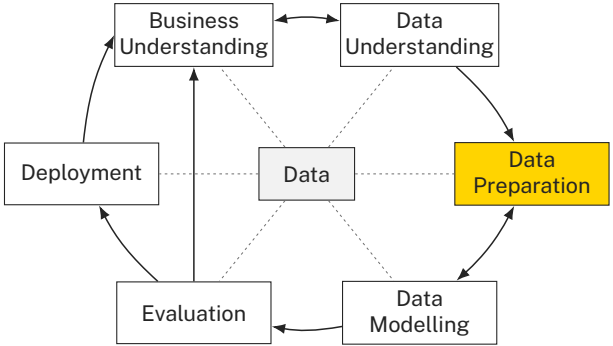

Cycle after the CRISP-DM reference model, Chapman et al. (2000).

# Data Preparation

## 3. Data Preparation

How do we clean and transform data for modelling?

Document changes to units, missing values, and selected rows.



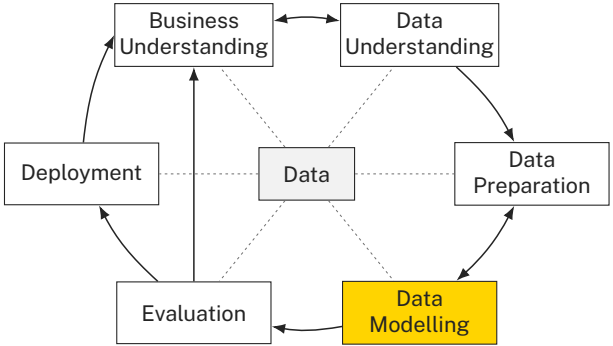

Cycle after the CRISP-DM reference model, Chapman et al. (2000).

# Data Modelling

## 4. Data Modelling

What modelling techniques should we use?

A method may require a different representation of the data. The arrow back to preparation records that dependency.



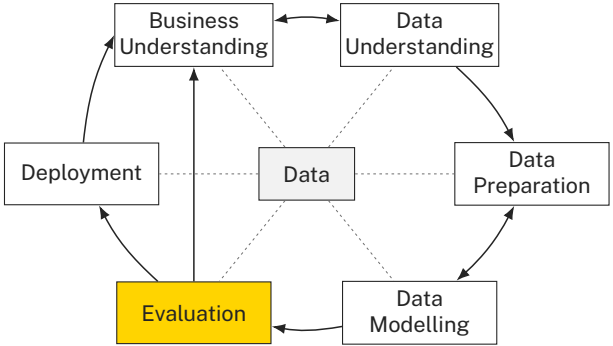

Cycle after the CRISP-DM reference model, Chapman et al. (2000).

# Evaluation

## 5. Evaluation

Does the selected solution meet the business goals?

This is a business question, not only an accuracy number. A model can score well and still fail the goal set in stage one.



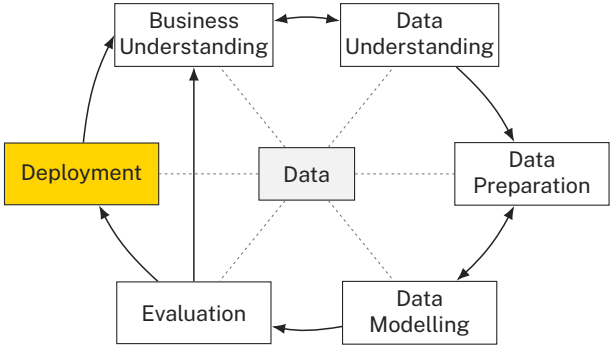

Cycle after the CRISP-DM reference model, Chapman et al. (2000).

# Deployment

## 6. Deployment

How do we deliver our solution to the business?

*What kind of deliverables would provide value to stakeholders?*




Cycle after the CRISP-DM reference model, Chapman et al. (2000).

ACTIVITY

# Think, Pair, Share

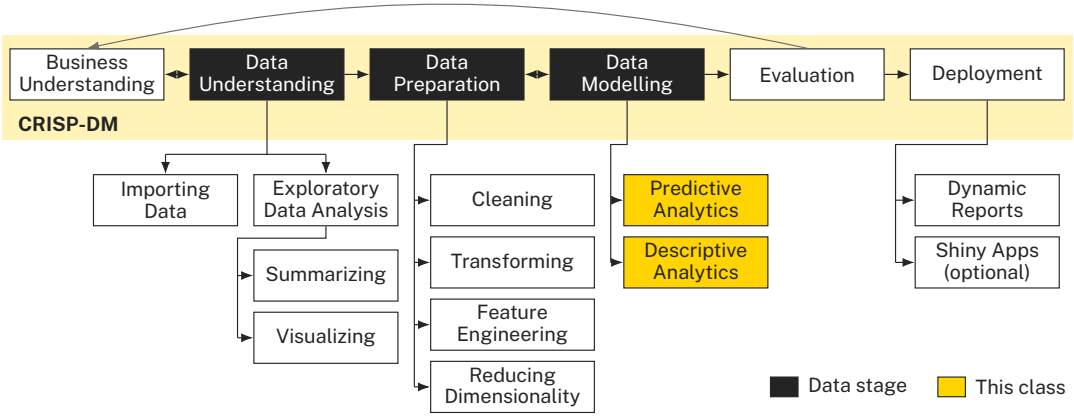
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1. Take a minute to think of a data science application you are curious about.
   - What data would you need? Do you think it would be readily available?
   - What challenges do you expect to face?
2. Discuss with your neighbour.
3. Volunteer to share with the class.

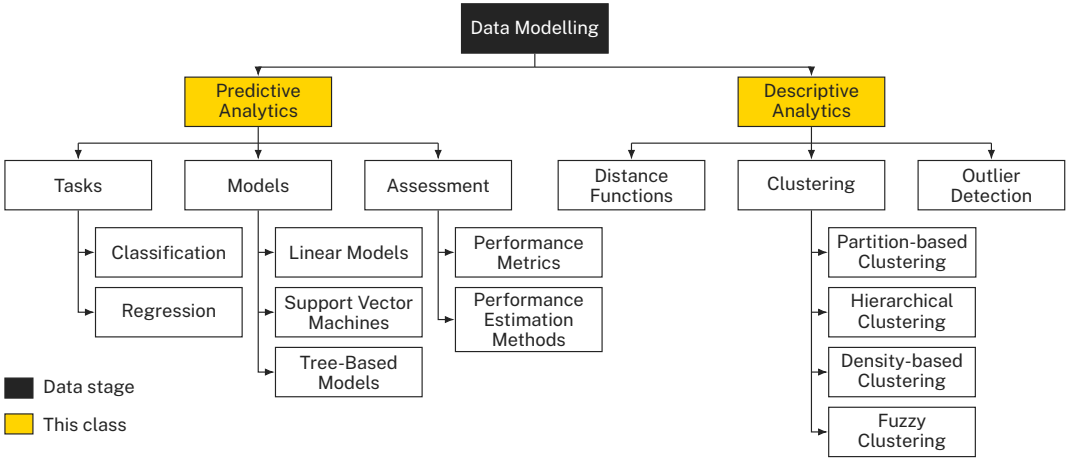
## About the Course

The syllabus is organized around the same stages, starting with data understanding and preparation.

# The Process Spine and the Data Stages



# The Modelling Branch



# A Working Environment for Thursday

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## The R Toolchain

- Use **Posit Cloud** or an R notebook on Kaggle.
- For local work, install **R** from CRAN and an editor such as **RStudio**.
- Week 1 uses base R. Install packages when a class introduces them.

## The Python Toolchain

- Use **Google Colab** or a Python notebook on Kaggle.
- For local work, use **Python 3**, a virtual environment, and an editor.
- Week 1 uses `pandas` for tables and `matplotlib` for plots.

Bring one working local or cloud environment. Both languages use the same rubric. Contact the instructor early if computer access is a barrier.

`cran.r-project.org` `posit.co/download/rstudio-desktop` `python.org/downloads` `code.visualstudio.com`

# Choose One Free Notebook Environment

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| Service | Language | Environment |
|---|---|---|
| Posit Cloud | R or Python | RStudio or Jupyter projects |
| Kaggle Notebooks | R or Python | Browser notebooks |
| Google Colab | Python | Browser notebooks |

- Local R or Python is equally acceptable. No paid service is required.
- Free services have usage limits. Save your files and record dependencies.
- Keep assessed notebooks private until the deadline and release to the class.
- No paid textbook is required. The repository's slides, notes, and lab handouts are the primary source; cheatsheets come with each module.

`posit.cloud` `kaggle.com/code` `colab.research.google.com`

# How We Stay in Contact

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- Announcements go in the course GitHub repository.
  - Activate e-mail notifications.
  - Check your Dalhousie e-mail daily.
- Use Teams for class questions and to arrange office hours.

## For Private Matters, Use E-mail

- Subject line: `[CSCI 3141] Subject matter`.
- Instructor: Prof. Gabriel Spadon, `spadon@dal.ca`.
- Replies normally take three business days.

Course repository, `github.com/MAPS-Lab/CSCI-3141`

# How Office Hours Work

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### By Appointment

Request a time through Microsoft Teams; there is no fixed weekly slot.

### Individually or in Small Groups

Ask for a group appointment when several of you share a question.

### Short and Focused

Say what you want to cover when you write, so the meeting can begin with your question.

### Ask Early

Leave time to use the feedback before your deadline.

# Academic Integrity and Conduct

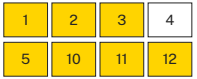
- Academic integrity is expected from all students at Dalhousie.
- Review the Academic Integrity resources in the course repository.
- Originality checking may be used, and follows syllabus A.9.1.
- Everyone at Dalhousie should be treated with dignity and respect.
- Keep this in mind in every interaction, in class and online.

**Generative AI** is prohibited on integrated labs and examinations.
Assignment use requires express authorization in its instructions.

Dalhousie academic integrity, `dal.ca/about/leadership-governance/academic-integrity.html`

# Integrated Labs (15%)

- Each lab applies the week's method and carries earlier decisions forward.
- One rubric covers the data, methods, interpretation, and reproducibility.
- The **best 7 of 8** lab results count, so one can be missed without penalty.
- Submit **seven calendar days** later, before **16:05 Halifax time**, even when no class meets that week.




Eight Offered, Best Seven Counted

Example: the lowest (4) drops; Assignment 2 covers Weeks 06 to 09

# Assignments (45%)

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| Assignment | Weight | What It Asks For | Due |
|---|---|---|---|
| Assignment 1 | **7.5%** | Import data and perform basic data exploration | 20 October 2026 |
| Assignment 2 | **7.5%** | Compare predictive models honestly | 16 November 2026 |
| Assignment 3 | **15%** | Design and implement a predictive modelling solution | 29 November 2026 |
| Assignment 4 | **15%** | Design and implement a descriptive modelling solution | 9 December 2026 |

The four assignments trace the syllabus arc, from import to prepare to predict to describe.

# More on Assignments

- Assignments are **individual** work.
- They are due at 23:59 on the announced date.

## Late Submissions

- Assignments 1 to 3 lose **5 percentage points per late calendar day**, up to seven days.
- Later work requires an approved absence or accommodation process.
- Assignment 4 has no routine late window.

Student Declaration of Absence and accommodation provisions apply as the University regulations set them out.

# From Private Work to Public Portfolio

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**1. Work privately**

Use a private repository with instructor access. Commit regularly. Keep your work private.

**2. Submit by the deadline**

Record the full commit SHA and commit URL in that directory's `README.md` and push it. That record fixes the assessed version.

**3. Share after release**

After the deadline and when the instructor opens sharing, publish that same commit and its history. Share it at the next class meeting, if one follows.

A branch in a public repository is public too. Use the syllabus's private-portfolio alternative when required.

# Keep the Evidence and Credit the Source

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**Preserve assessment history**

- Keep the submitted commit reachable permanently.
- No rewriting, squashing, deletion, force pushes, or backdating.
- Make corrections in new commits. Git dates alone do not prove timely submission.

**Share released materials**

- Credit the author and CSCI 3141. Preserve source links and licence terms.
- Keep instructor-marked private material, grades, and feedback private.
- Do not share assessed work early or share another student's work.

# Midterm and Final Examination (40%)

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## Midterm (15%)

- Held **in class** on October 20, 16:05 to 17:25.
- Covers everything up to that class.

## Final Examination (25%)

- Scheduled by the Registrar.
- **Cumulative**, covering everything.

Bring valid Dalhousie photo ID. Course and personal notes are allowed unless the examination lists narrower approved aids. Internet access, communication, and generative AI are prohibited.

# Grading Summary

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| Component | Weight | Rule |
|---|---|---|
| Integrated labs | **15%** | best 7 of 8 results |
| Assignment 1 | **7.5%** | individual |
| Assignment 2 | **7.5%** | individual |
| Assignment 3 | **15%** | individual |
| Assignment 4 | **15%** | individual |
| Midterm | **15%** | in class, 20 October 2026 |
| Final examination | **25%** | cumulative, Registrar-scheduled |
| Completion requirement | **Ungraded** | must be met to pass, syllabus A.7.4 |
| **Total** | **100%** | |

**The weighted total, the combined assignment component, and the final examination must each reach 50%, and the completion requirement in syllabus A.7.4 must be met.**

# Dates to Put in Your Calendar Now

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### Course Dates

| Event | Date |
|---|---|
| Classes begin | 8 Sep |
| Midterm, in class | **20 Oct** |
| Assignment 1 | 20 Oct |
| Assignment 2 | 16 Nov |
| Assignment 3 | 29 Nov |
| Last class | 3 Dec |
| Assignment 4 | **9 Dec** |

### University Deadlines

| Deadline | Date |
|---|---|
| Last day to add | 22 Sep |
| Drop without a W | **7 Oct** |
| Drop with a W | **5 Nov** |
| Fall Break | 9–13 Nov |
| Final examination | set by the Registrar |

The course GitHub repository carries the authoritative version if anything changes.

University dates, `dal.ca/study/plan-your-degree/important-dates.html`

SUMMARY

# What Today's Class Covered

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### Why Data Science

1

Data-related roles involve different work. This course begins with understanding the data itself.

### What Data Science Is

2

Learning from data through computing, statistics, and domain knowledge. This course focuses on tables.

### Three Kinds of Question

3

Exploratory, predictive, and descriptive, asked of one Heart Disease table inside the CRISP-DM loop.

### How This Course Works

4

Best 7 of 8 labs, four assignments, a midterm, a cumulative final, and the A.7.4 completion requirement.